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RESEARCH-ARTICLE

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HAIFENG WEI, Hefei Technology College, Hefei, Anhui, China

Open Access Support provided by:
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Published: 30 May 2024

[Citation in BibTeX format](#)

MIDA 2024: 2024 International
Conference on Machine Intelligence and
Digital Applications
May 30 - 31, 2024
Ningbo, China

Intelligent Legal Document Generation System and Method Based on Knowledge Graph

Haifeng Wei

Hefei Technology College, Hefei, Anhui, 238000, China
weihf880@163.com

ABSTRACT

Applying artificial intelligence to legal research will play a very important role in improving judicial efficiency and judicial quality. In order to solve the problems existing in the current automatic generation of judicial documents, such as loose logic, incomplete content and irregular language specifications, an intelligent automatic generation method of judicial documents based on knowledge graphs was applied. Based on the knowledge graph in the legal field, it uses its powerful reasoning and correlation to realize the intelligent automatic generation of legal issues. The study found that this method is superior to conventional rule-based search methods in multiple dimensions such as generation efficiency, quality, and content integrity, and its content integrity can reach up to 92%. This method can effectively conduct efficient and stable text queries, greatly improving the automation level of legal document generation. The application of knowledge graphs to legal research will greatly improve the efficiency and quality of trial document production, and is expected to bring about revolutionary changes in legal research and documentation, and promote the simplification and precision of the litigation process.

CCS CONCEPTS

• **Social and professional topics** → Professional topics; Management of computing and information systems.

KEYWORDS

Knowledge Graph, Intelligent Law, Document Generation, Content Integrity

ACM Reference Format:

Haifeng Wei. 2024. Intelligent Legal Document Generation System and Method Based on Knowledge Graph. In *2024 International Conference on Machine Intelligence and Digital Applications (MIDA 2024)*, May 30, 31, 2024, Ningbo, China. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3662739.3669909>

1 INTRODUCTION

In the current process of judicial informatization, the development and advancement of trial work has made the trial work more efficient and transparent. With the rapid development of artificial

intelligence, especially its application in legal research, a new way of thinking has been proposed to improve the efficiency and quality of trial work [1, 2]. System changes such as smart trial, smart trial, and smart delivery have greatly improved the efficiency and quality of trial work. Judicial decision documents are the main medium of judicial power, and their degree of automation is of great significance to judicial decisions [3, 4]. However, the current automatic generation system of judicial documents has logical flaws, incomplete content, and irregular language, and needs to be further improved.

There have been some studies on the automatic generation of legal documents. Among them, Anand uses deep learning methods to effectively summarize legal text summaries [5]. For specific legal documents, Vaissnave has designed a variety of legal document automatic generation systems based on automatic firefly swarm optimization deep learning models, such as rule-based automatic generation systems, statistics-based automatic generation systems and machine learning-based automatic generation systems. etc.[6]. Through an extensive review of the current status of this research field, this paper compares existing rule-based automatic generation systems, statistics-based automatic generation systems, and machine learning-based automatic generation systems, which show significant shortcomings in accuracy, standardization, and consistency. In order to solve the above problems, this article studies an intelligent automatic generation system of judicial documents for judicial cases and its key technologies.

It is composed of relationships and attributes, and builds a knowledge network through the relationships between entities. In the judicial field, entities can refer to the individuals in the case, the cause of the case, as well as the object, time, location and other information of the case. During the trial, they can be the person in charge of the case, members of the collegial panel, or the office address of the court [7, 8]. By mapping entities and their relationships onto knowledge graphs, the problems of data asymmetry and low quality in the legal field can be solved to a certain extent. The intelligent legal document generation system studied in this article mainly uses a knowledge graph-based method to map the relationships between entities onto the knowledge graph [9, 10]. It utilizes extractive neural networks and relationship extraction technology to extract case information and enable automatic document generation.

By establishing a legal corpus database, the massive knowledge in legal texts can be automatically extracted. Utilizing the concept and logic of "knowledge map", the unified definition and management of case topics are achieved. Through the semantic understanding of case topics and the calculation of deep semantic similarities, the case topics are correlated and compared, so as to achieve the intelligent automatic generation of court documents [7, 8]. Compared

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MIDA 2024, May 30, 31, 2024, Ningbo, China

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ACM ISBN 979-8-4007-1814-4/24/05
<https://doi.org/10.1145/3662739.3669909>

with the traditional rule-based retrieval algorithm, this algorithm not only improves the retrieval speed, but also improves the retrieval quality. Experimental data shows that this algorithm can significantly improve the overall quality of the full text, improve its standardization level, and reduce the overall absolute error rate. It has great practical significance for promoting the development of judicial document automation technology and improving the efficiency and quality of judicial case handling [9, 10]. And its in-depth application in the field of law will be discussed and improved to meet the growing judicial needs.

2 INTELLIGENT LEGAL DOCUMENT GENERATION SYSTEM UNDER KNOWLEDGE GRAPH

2.1 Principles and construction of knowledge graph

This paper adopts a new research idea, which is to use machine learning technology to achieve automatic recognition and establishment of text targets. Use deep convolutional neural networks and recurrent neural networks, especially long short-term memory networks and gated recurrent units, to achieve effective recognition and classification of names, locations, organizations, etc. in text. For relationship extraction, we use guided learning methods based on support vector machines, neural networks, etc. to extract semantic associations containing multiple words such as "belongs to" and "located in" from the labeled data. This method can extract meaningful information from the data to construct connections between objects. Using the idea of ontology disambiguation, the clustering algorithm is combined with cosine similarity to achieve effective extraction and fusion of multiple quotes of the same object. This process ensures correctness and consistency in the knowledge base [11, 12]. For specific technical problems, this article uses specific modeling methods and calculation methods to optimize the algorithm. Taking CNN as an example, we use convolution and precipitation layers to extract features, the formula is expressed as:

$$f(x) = \text{ReLU}(W \times x + b) \quad (1)$$

Where W is the convolution kernel, x is the input feature map, b is the bias term, and ReLU is the activation function. Through these methods, a system can be built that can automatically extract information from text and build a knowledge graph.

2.2 Relationship extraction

In order to improve the automatic recognition ability of targets in judicial documents, this project proposes a dictionary-based retrieval method. This process relies on guided learning, such as support vector machines, neural networks, etc., to identify semantic connections between entities through training on labeled data. One of the important issues is to determine whether multiple entities mentioned refer to the same real-world object, and to ensure that they are consistent and correct. On this basis, CBoW is used to optimize relationship extraction. The CBoW model predicts the occurrence probability of words by judging contextual conditions, and corrects the association between words. The target formula is

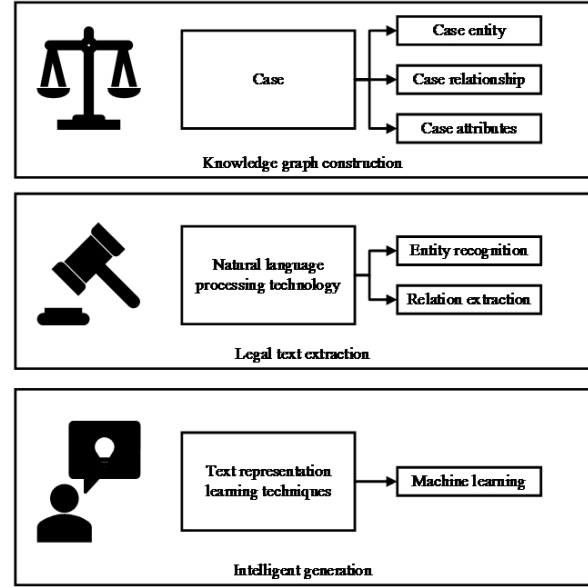


Figure 1: Design framework of intelligent legal document generation system

as follows:

$$\prod_{t=1}^T p(w_t | \text{Context}(w_t)) = \prod_{t=1}^T p(w_t | w_{t-c}, \dots, w_{t-1}, w_{t+1}, \dots, w_{t+c}) \quad (2)$$

Among them, w_t is the central word, w_{t-1} , w_{t+1} are the top and bottom words, T represents the total vocabulary of the text, and c represents the window size for predicting word context. The choice of window size c has a significant impact on model performance. A smaller window can capture more specific background information, but it can miss some helpful context. A larger window can provide more contextual information, but will also produce some words that are semantically irrelevant. At the same time, the objective function can also be expressed by the following formula:

$$L = \sum_{t=1}^T \log p(w_t | \text{Context}(w_t)) \quad (3)$$

The window size based on CBoW mode determines the expected contextual area of the vocabulary. This article will use word vectors to express context, use existing word vectors to describe words, convert them into vectors of a certain length that contain semantic connections between words, and average the word vectors obtained after weighting them, so as to Get the context vector that represents the context in which the word is located.

2.3 System design

The intelligent legal document generation method based on knowledge graph proposed in this article is based on the classification of legal documents. For different types of cases, relevant information is extracted by classifying the document data, and then relevant documents are automatically generated. The system design framework is shown in Figure 1.

This article focuses on examples and integrates the entities, relationships, and attributes in the examples. By performing operations such as word segmentation, part-of-speech tagging, and grammatical analysis on documents, quantitative expression of document data is achieved. Methods such as text representation and machine learning are used to extract features from the collected legal documents to achieve automation. File processing technology divides legal documents into several independent word units, and assigns corresponding part-of-speech tags to each word, such as nouns, verbs, etc [13, 14]. Then analyze the syntactic structure and judge their interconnections. Through the above three links, the transformation from unstructured text to structured text is realized, and a good foundation is laid for future feature extraction and file creation.

In automatic text generation, feature extraction is an important link. First, judicial documents are converted into a data form that can be modeled by machine learning, and then machine learning methods such as decision trees, random forests, and gradient boosting machines are used to extract the documents. In the process of establishing ontology, this article defines the following concepts: Entity: such as "person", "object", "place", "organization", etc. Relationship: such as "reason", "include", "inherit", etc. This concept helps to build a knowledge graph that can provide a structured knowledge base for document generation [15]. In order to achieve the above functions, this article uses the conditional random field formula:

$$P(y|x) = \frac{1}{Z(x)} \prod_{i=1}^N P(y_i|y_{i-1}, x_i) \quad (4)$$

$P(y|x)$ is the probability of label sequence y given input x , and $Z(x)$ is the normalization factor.

2.4 Legal text extraction

This article combines knowledge graph construction with natural language processing technology, using natural language processing technology to process legal texts such as word segmentation, part of speech tagging, syntactic analysis, etc., and using the CBoW model to extract features from document data. This method adopts a combination of rule-based and machine learning based methods, utilizing natural language processing techniques to convert legal texts into digital form, that is, by processing document data through word segmentation, part of speech tagging, syntactic analysis, and other processes. It then converts it into digital vector form, and finally uses the Bidirectional Encoder Representations from Transformers (BERT) model to extract features from the digital vector, thereby generating intelligent legal documents. The combination of rule-based and machine learning based methods adopted in this article can effectively improve the accuracy of legal text classification [16, 17].

In the process of association extraction, the CBoW-based concept is transformed from the vector expression of a single word into the association between two entities. The learning process of this method is as follows: First, collect and pre-process legal text, including segmentation, deletion of pause points, and extraction of stems. On this basis, CBoW-based vocabulary vectors are used to extract associations; then, association classes are defined in the training samples and labeled; finally, using samples with existing labels, a model for identifying objects is established classifier for

correlation. In the legal field, natural language processing (NLP) is an important means to achieve legal vocabulary recognition, legal concept extraction and legal knowledge base. On this basis, a method based on semantic analysis is proposed to realize the analysis of entities and their correlations in legal texts, laying the foundation for establishing a semantic-based knowledge graph.

3 EXPERIMENTAL DESIGN AND DATA

3.1 Experimental design

The test dataset used in this article is the court document dataset. The basic test dataset selected in this article mainly includes the judicial judgment retrieval dataset and judgment corpus data, as well as the judicial judgment retrieval dataset, including case names, case numbers, court names, judge names, and judgment results. The judgment corpus dataset mainly includes the textual content of various judgment documents, including the court's trial process of the case, the citation of relevant legal provisions, the judge's interpretation and judgment of the case, and other specific content. The training set used in this article is judicial judgment texts, and the testing set is judicial document corpus. The legal knowledge graph data from the knowledge graph service platform used in the experimental process of this article. It compares the knowledge graph based intelligent legal document generation system designed in this article with the rule-based methods in existing intelligent legal document generation systems. This article analyzes the generation efficiency, quality, content integrity, and language standardization of the two in the process of intelligent legal document generation to evaluate the effectiveness of the system application. This article compares experimental data on different types of cases and evaluates the stability of system applications by calculating the average absolute error of the relevant data. This article would conduct experiments on the application of these two systems with different types of cases, as shown in Table 1.

3.2 Generation efficiency

Based on the above experimental design, the generation efficiency data obtained in this article is shown in Figure 2.

From Figure 2, it can be seen that rule-based methods have lower generation efficiency than knowledge graph methods, and their efficiency is significantly lower in different types of cases. The rule-based method takes up to 45 minutes to generate a legal document, while the knowledge graph method only takes up to 30 minutes. In addition, the knowledge graph method can generate an article in 15 minutes at the fastest, while the rule-based method can only generate an article in 25 minutes at the fastest. The knowledge graph method has higher efficiency and faster generation speed in generating legal documents. This is very beneficial for improving the efficiency of legal document generation and reducing time costs. The knowledge graph method has greater flexibility and applicability when processing different types of legal documents, can more quickly adapt to the needs of different case types, and improves production efficiency.

Table 1: Classification of case types

Case type	Case description	Caseload
Case for crown	Cases involving criminal offenses	2
Civil case	Civil disputes and contract disputes and other civil cases+	2
Administrative case	Administrative act disputes and administrative adjudication and other cases	2
Intellectual property cases	Patent, trademark, copyright and other intellectual property rights cases	2
Divorce case	Family divorce and child support cases	2

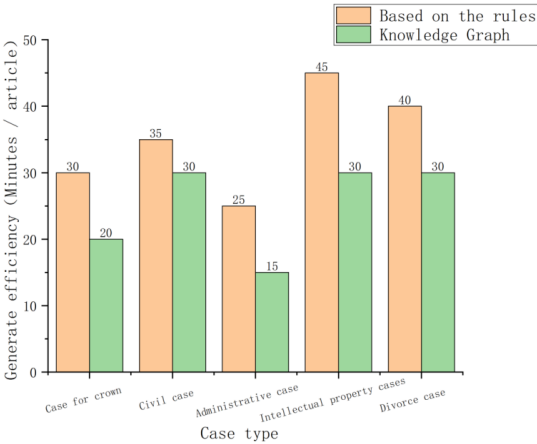


Figure 2: Generation efficiency

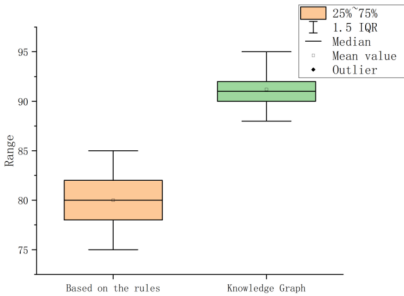


Figure 3: Generation quality

3.3 Generation quality

In terms of generation quality, this article would score the generation quality with a value of 1-100, and the scoring results are shown in Figure 3.

From Figure 3, it can be seen that the generation quality of traditional rule-based methods ranges from 75 to 85, while the generation quality of knowledge graph based methods can reach a range of 88 to 95. From the perspective of mean, the traditional rule-based method has a mean of 80, while the knowledge graph based method can achieve a mean of 91.2. In terms of the quality of generated documents, knowledge graph-based methods have

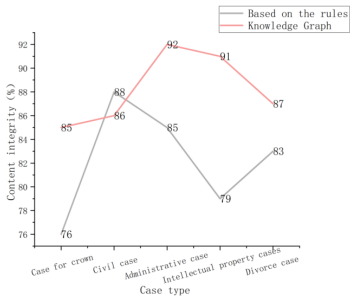


Figure 4: Content integrity

Table 2: Average absolute error data

Evaluation indicators	Based on the rules	Knowledge graph
Generate efficiency	6	6
Generate quality	2.8	1.84
Content integrity	3.76	2.64

higher average quality than traditional rule-based methods, and are more stable and superior in quality range.

3.4 Content integrity

The content integrity data collected in this article is shown in Figure 4

The data in Figure 4 shows that the content integrity of knowledge graph based methods ranges from 85% to 92%, with a maximum of 92%, while the content integrity of rule-based methods ranges from 76% to 88%. Moreover, the data of the knowledge graph based method shown in Figure 4 is smoother, indicating that the knowledge graph based method is more stable in terms of content integrity. In the handling of civil cases, the content integrity of knowledge graphs is lower than that of rule-based methods. Overall, the content integrity of the knowledge graph is higher, and the knowledge graph method can more stably ensure the integrity of the content of the generated legal documents, which is of great significance for ensuring the quality of the generated documents.

3.5 Average absolute error

Based on the above data, this article calculates the average absolute error and obtains the data as shown in Table 2.

The average absolute error of rule-based methods and knowledge graph methods is the same in terms of generation efficiency, but rule-based methods have a larger average absolute error in terms

of generation quality and content integrity. Compared with traditional methods, the method proposed in this article can achieve an average absolute error reduction of 34.3% and 29.8% in generation quality and content integrity. Traditional methods are not as stable as knowledge graph methods in terms of generation quality and content integrity. Although the knowledge graph algorithm has obvious advantages in generating legal documents at this stage, its application in the legal field is far from saturated, and there are a large number of potential application scenarios waiting to be explored. You can try to introduce more advanced information retrieval and natural language processing technologies to improve the system's retrieval accuracy and generation efficiency.

4 CONCLUSIONS

This paper designs an intelligent legal document generation system based on knowledge graph. Compared with the rule-based search method, this method shows obvious advantages in terms of generation efficiency, generation quality and content completeness. The study found that this method can achieve efficient and reliable document retrieval, thus improving the degree of document automation. This paper innovatively builds a case-based knowledge graph, which not only contains the ontology and attributes of the instances, but also reflects the complex relationships between instances, and can provide a large amount of semantic information for the automated creation of documents. Future research can further expand its application in legal research, including the expansion of legal knowledge graphs, the integration of multi-modal legal data, and the automatic generation of legal documents.

REFERENCES

- [1] Pagallo U, Ciani Sciollo J, Durante M. The environmental challenges of AI in EU law: lessons learned from the Artificial Intelligence Act (AIA) with its drawbacks. Transforming Government: People, Process and Policy, 2022, 16(3): 359-376.
- [2] Roy P, Bag S. Ink analysis based forensic investigation of handwritten legal documents. Multimedia Tools and Applications, 2022, 81(16): 23007-23047.
- [3] An ZW, Lai YX, Feng YS. Natural language understanding for legal documents. Journal of Chinese Information Science, 2022, 36(8): 1-11.
- [4] Wei B, Kuang K, Sun C. A full-process intelligent trial system for smart courts. Frontiers, 2022, 23(2): 186-206.
- [5] Anand D, Wagh R. Effective deep learning approaches for summarization of legal texts. Journal of King Saud University-Computer and Information Sciences, 2022, 34(5): 2141-2150.
- [6] Vaissnave V, Deepalakshmi P. Modeling of automated glowworm swarm optimization based deep learning model for legal text summarization. Multimedia Tools and Applications, 2023, 82(11): 17175-17194.
- [7] Hogan A, Blomqvist E, Cochez M. Knowledge graphs. ACM Computing Surveys (Csur), 2021, 54(4): 1-37.
- [8] Sun ZL, Hu JH. Problems and solutions when applying artificial intelligence in judicial trials. Journal of Zhejiang University of Technology (Social Science Edition), 2021, 20(4): 405-411.
- [9] Liu Q, Yu ZT, Gao SX. Similar case matching integrating case elements. Journal of Chinese Information Science, 2022, 36(11): 140-147.
- [10] Li Z, Liu H, Zhang Z. Learning knowledge graph embedding with heterogeneous relation attention networks. IEEE Transactions on Neural Networks and Learning Systems, 2021, 33(8): 3961-3973.
- [11] Wang Y, Lipka N, Rossi R A, et al. Knowledge graph prompting for multi-document question answering. Proceedings of the AAAI Conference on Artificial Intelligence, 2024, 38(17): 19206-19214.
- [12] Zhou B, Bao J, Chen Z, et al. KGAssembly: Knowledge graph-driven assembly process generation and evaluation for complex components. International Journal of Computer Integrated Manufacturing, 2022, 35(10-11): 1151-1171.
- [13] Paul S, Goyal P, Ghosh S. LeSICiN: a heterogeneous graph-based approach for automatic legal statute identification from Indian legal documents. Proceedings of the AAAI conference on artificial intelligence, 2022, 36(10): 11139-11146.
- [14] Yang B, Liao Y. Research on enterprise risk knowledge graph based on multi-source data fusion Neural Computing and Applications, 2022, 34(4): 2569-2582.
- [15] Zhou B, Shen X, Lu Y, et al. Semantic-aware event link reasoning over industrial knowledge graph embedding time series data. International Journal of Production Research, 2023, 61(12): 4117-4134.
- [16] Chaudhri V, Baru C, Chittar N, et al. Knowledge graphs: introduction, history and perspectives. AI Magazine, 2022, 43(1): 17-29.
- [17] Gaur M, Gunaratna K, Srinivasan V, et al. Iseek: Information seeking question generation using dynamic meta-information retrieval and knowledge graphs. Proceedings of the AAAI Conference on Artificial Intelligence, 2022, 36(10): 10672-10680.